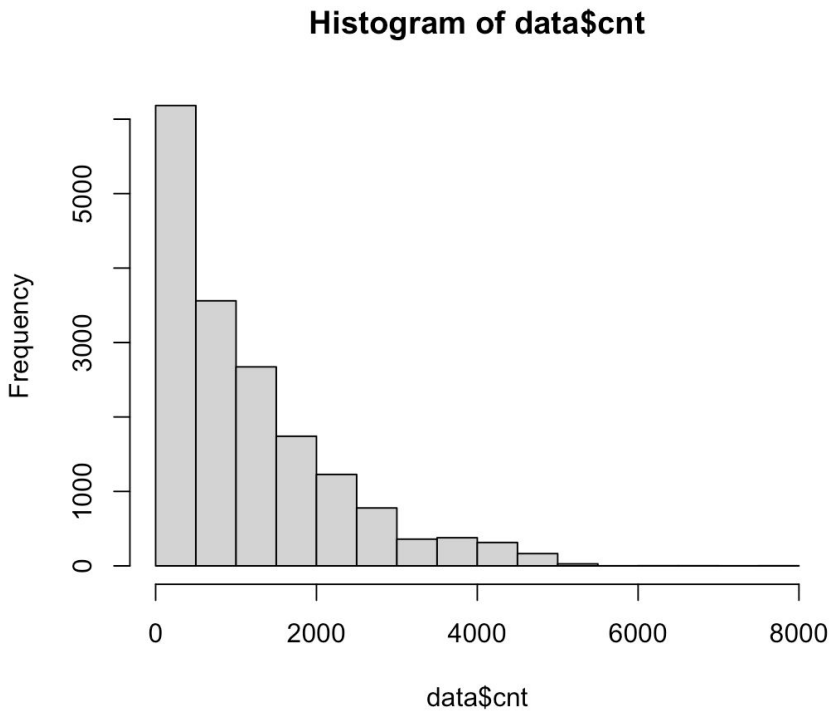
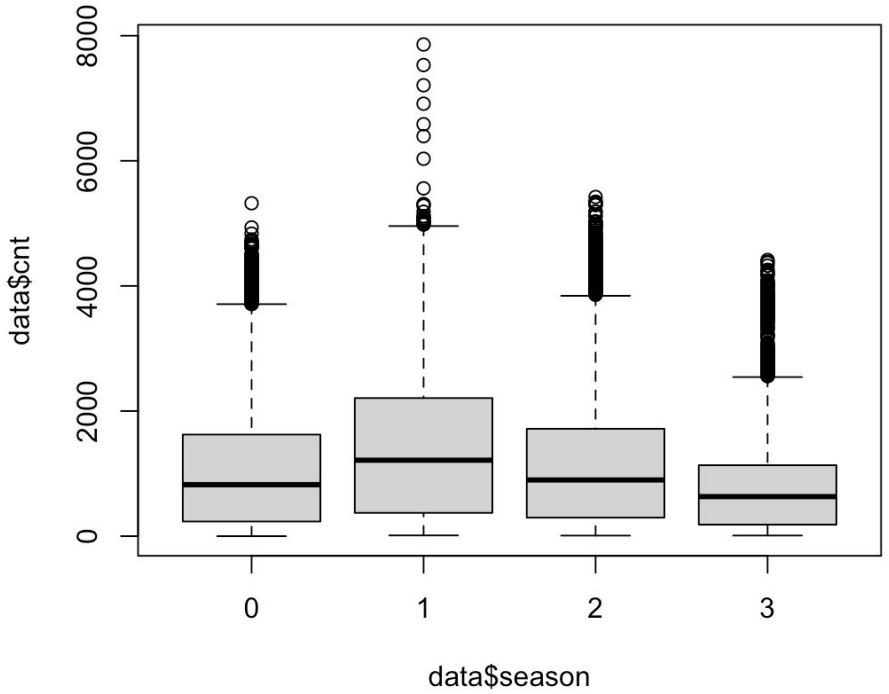
Bike sharing data in London

Histogram



Histogram of bike count. Plot shows that it is more common for a few number of bikes to be in use at any given time. Most common is 0-500 bikes in use.

Box plot of count vs seasons



Most common season to rent a bike is season 1= summer. The largest outliers are in the summer season and leads to the highest variability in number of bikes rented.

Extreme outliers, valid data or special event?

```
> bikeData[which(cnt>6000),]
              timestamp  cnt   t1   t2  hum wind_speed weather_code is_holiday is_weekend season
4462 2015-07-09 08:00:00 7531 14.5 14.5 61.0       19.0            2          0          0      1
4470 2015-07-09 16:00:00 6033 23.0 22.0 26.0       11.0            1          0          0      1
4471 2015-07-09 17:00:00 7860 23.0 22.0 27.0       11.0            1          0          0      1
4472 2015-07-09 18:00:00 6913 22.5 21.5 29.0       13.0            1          0          0      1
5130 2015-08-06 08:00:00 6585 19.0 19.0 78.0       12.0            7          0          0      1
5139 2015-08-06 17:00:00 7208 22.5 22.5 55.0       17.5            2          0          0      1
5140 2015-08-06 18:00:00 6394 21.5 21.5 58.5       20.0            2          0          0      1
```

London Tube strike: Service used by millions crippled

Updated 10:51 PM EDT, Thu July 9, 2015

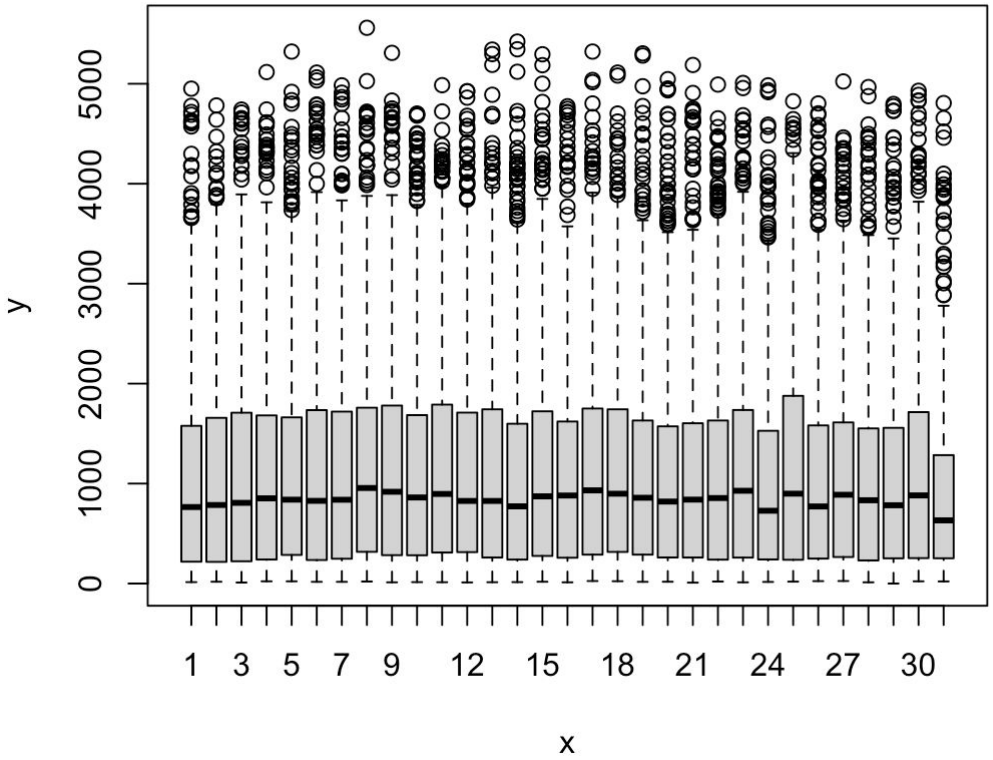


Extreme counts are observed during traffic heavy hours (going to or from work) during special days.

-> They are not representative to desired correlations.

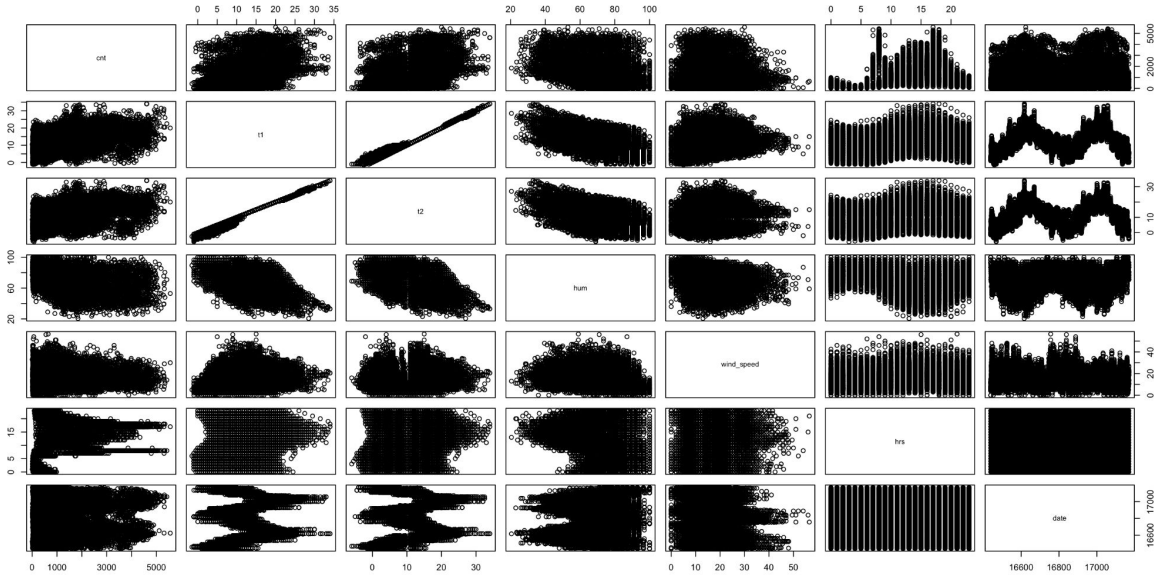
So in our future data we assume they are removed.

Count per day of month



There is little variation between the days. Outlier is the 31st since only half the months contain 31 days.

Pairwise comparison of continuous variables

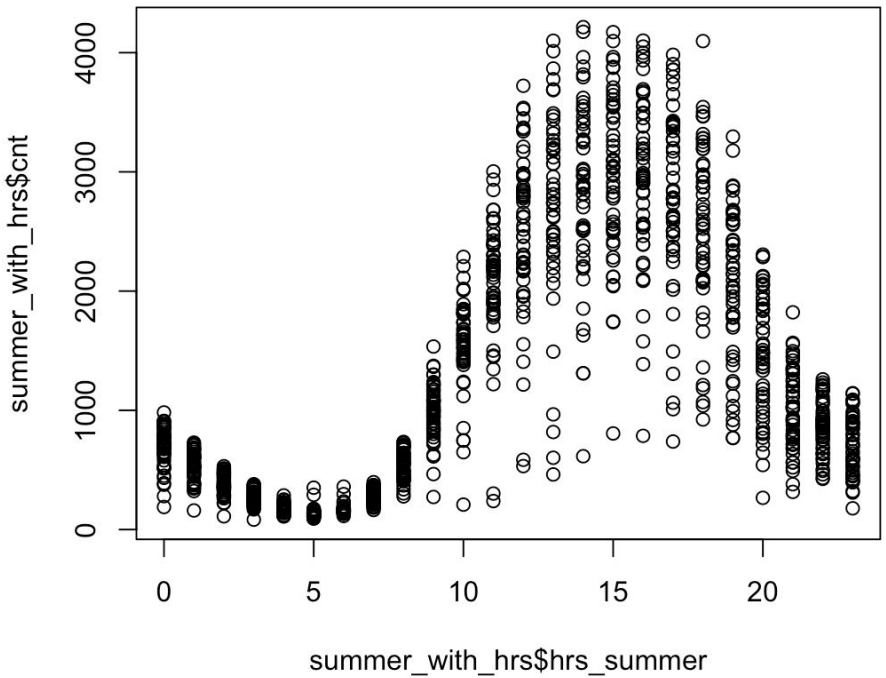


In the pairwise comparison between continuous variables we note that both t1 and t2 could be redundant as they exhibit a almost linear relationship.

Strong cyclic relationship between cnt vs hrs.

Cnt also seems to have a positive relation with t1 (or t2) but seems to be impacted by other variables as well.

Time of day vs count



Bike count during the hours of the day during summer, no holiday and the weekends.

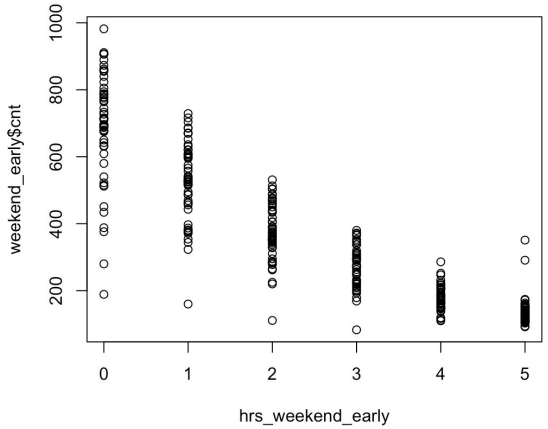
Clear trend of bikes rented at a given time otherwise. Can be candidate for covariance of count?

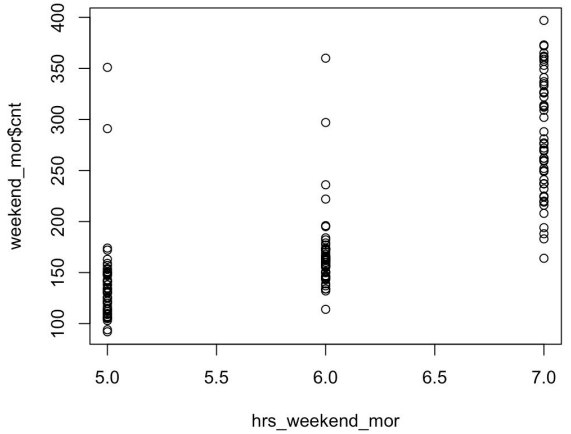
Could piecewise linear regression could do the trick?

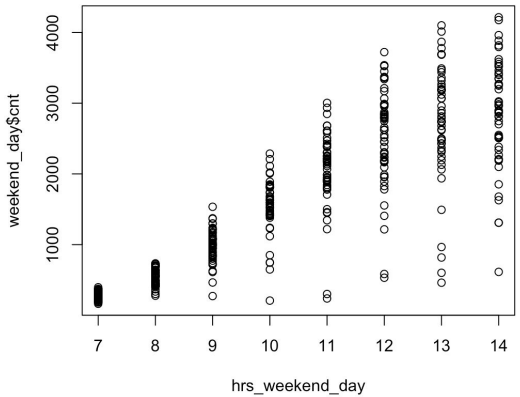
Dividing the time into 4 separate time subsets:

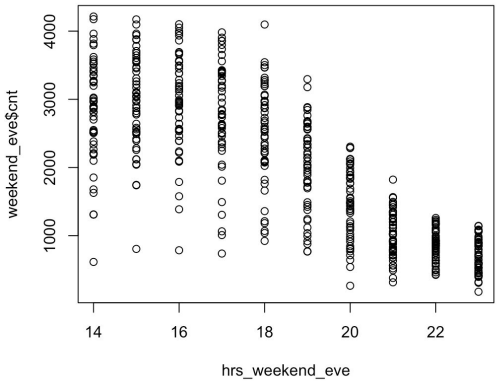
- 0 to 5 am
- 5 am to 9 am
- 10 am to 4 pm
- 5 pm to midnight

Linear regression might not be possible, use boxcox to transform data.









Lambda calc.

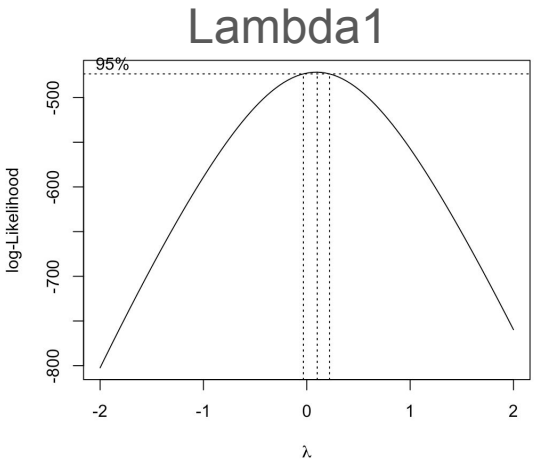
Box-Cox to transform data. Determine lambda for each time subset of data.

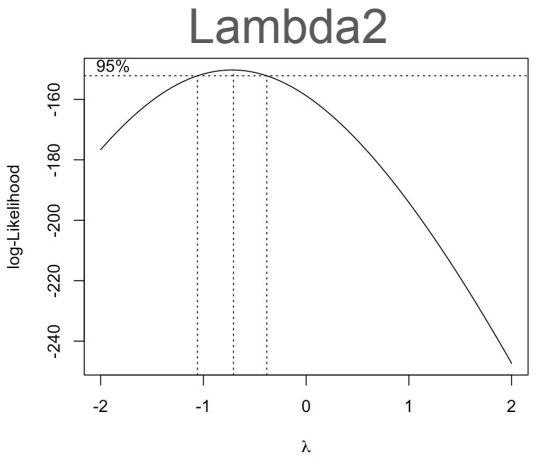
lambda1 = 0

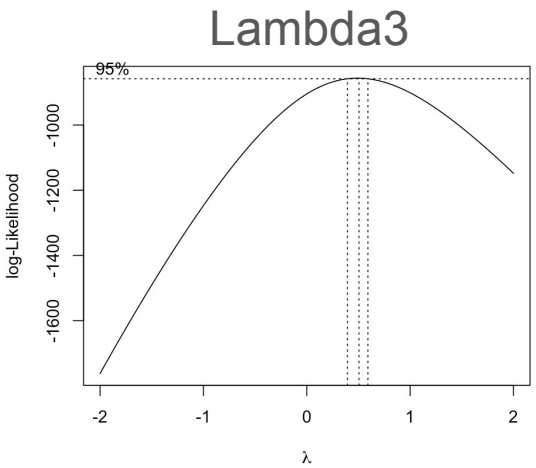
lambda2 = -0.707

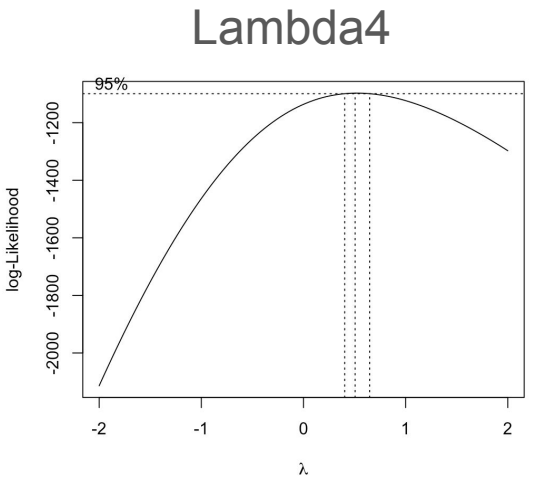
lambda3 = 0.505 (square root)

lambda4 = 0.505 (square root)





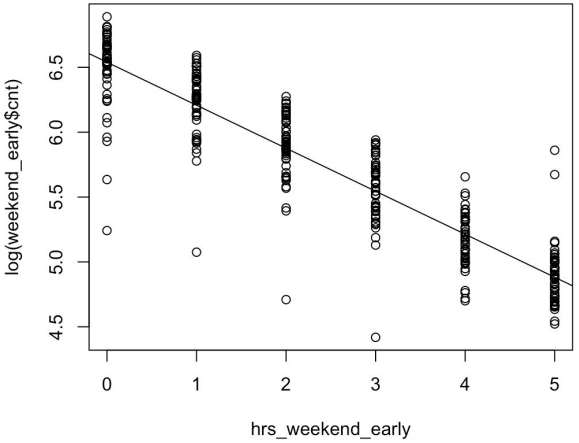


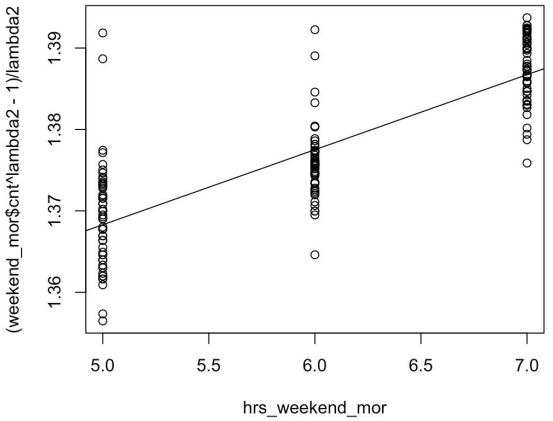


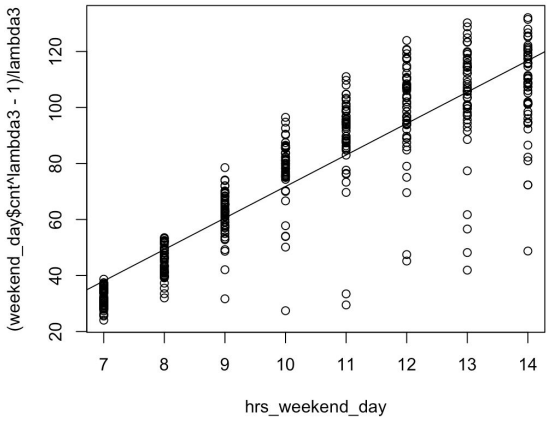
Box-Cox modified data with linear regression

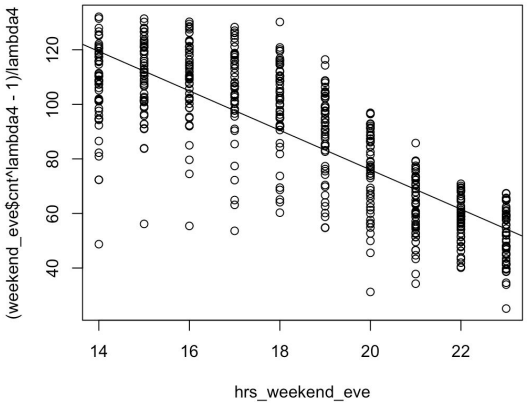
The data in each subset is modified according to Box-Cox with each respective lambda.

Linear regression line added to each subset.





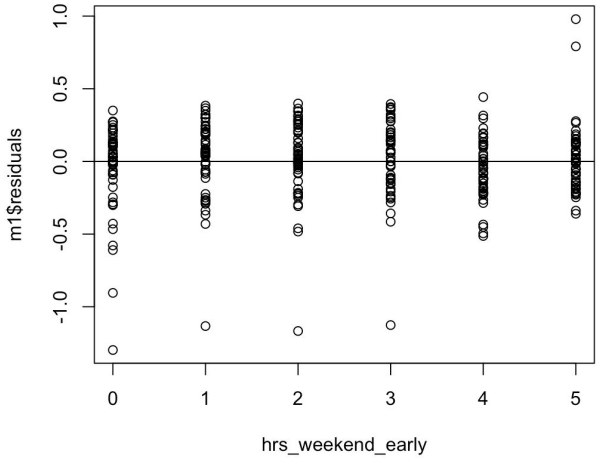


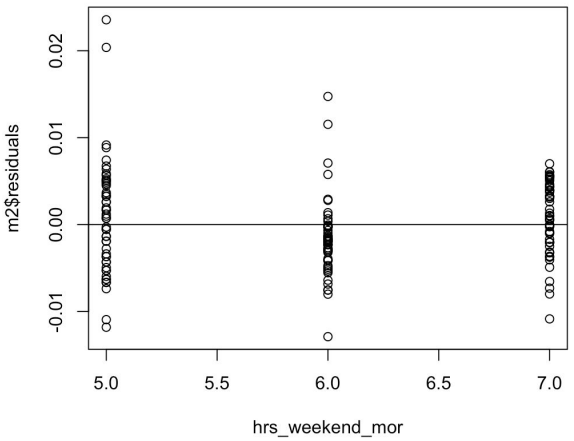


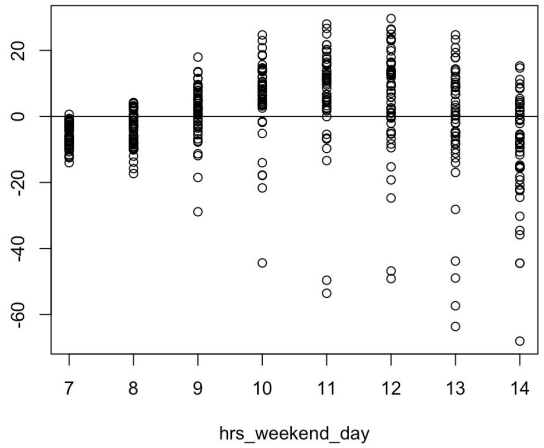
Residuals of linear regression

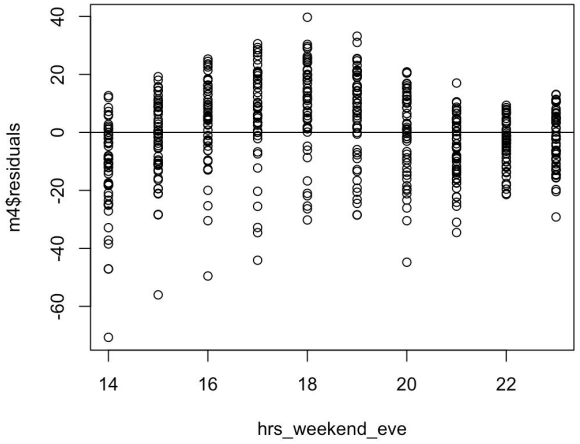
Residuals of each linear regression plot.

Regression looks ok for the early hours but not so good for the day and eve where there is a large variation









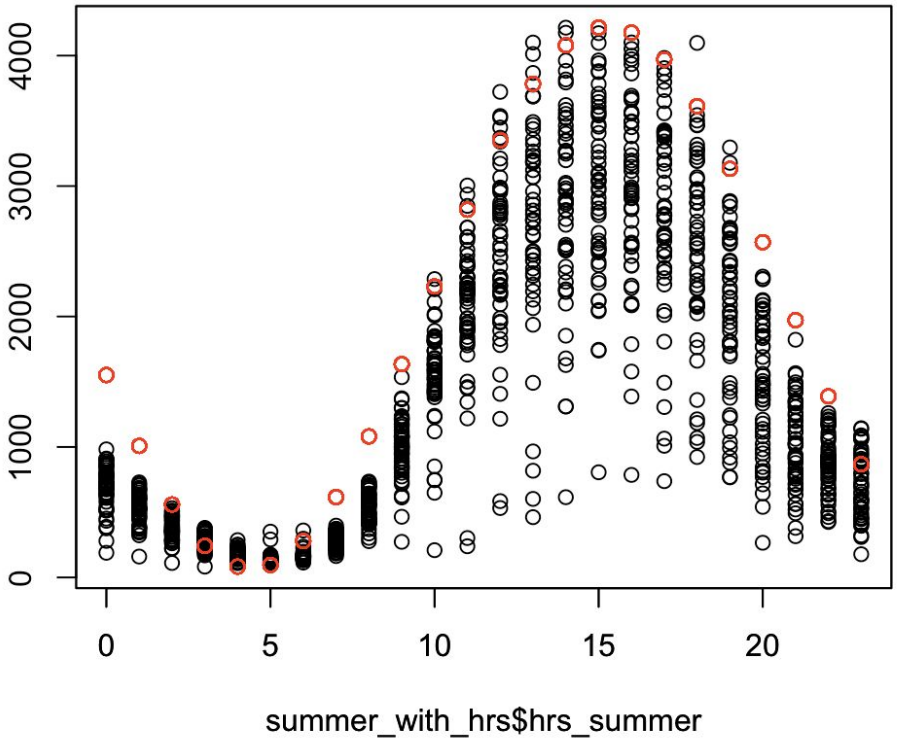
Interpreting Box-Cox transformation

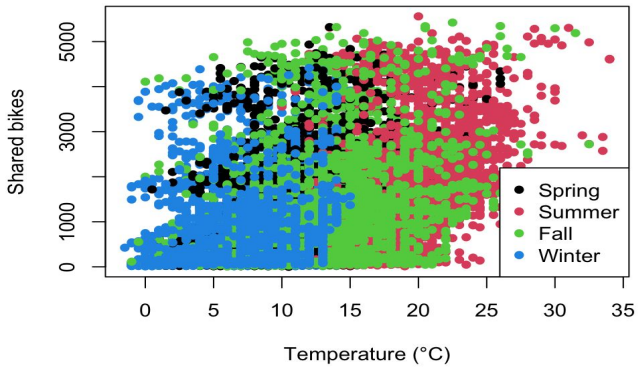
Looking at the four separate subsets, both transformations are used.

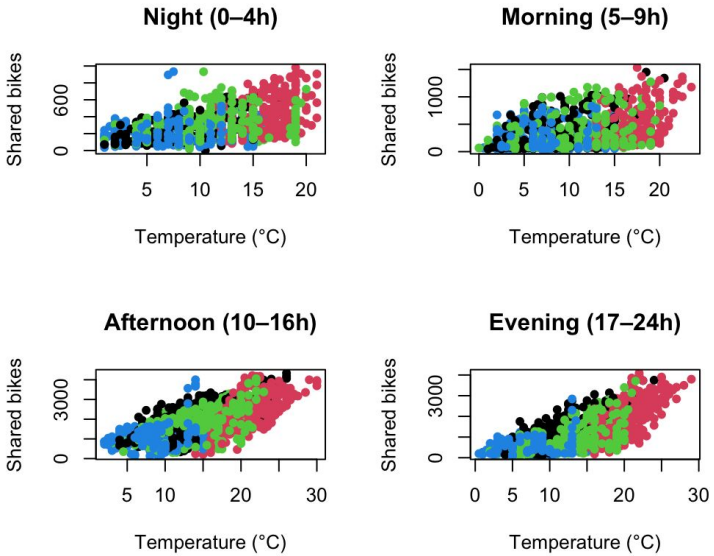
- The first subset stated lambda = 0 which means log(y) is used as the transformation and the linear regression fit well to the data. Explicitly stated we can say that when a hour elapses we expect the amount of shared bikes to decrease by 28%.
- The second subset stated lambda = -0.707 (-1 within the 95% confidence interval). The reciprocal (1/y) could be a good fit.
- The third and forth subset stated lambda = 0.505. Square root of y is a good fit for the data. Positive skewness of data when lambda < 1. This means the lower values of y will be stretched further down while the larger numbers will be pulled.

Sinusoidal curve for fitted data

Looking at the curve and the subsequent lambda from the subsets, a sinusoidal curve can better describe the entire series.







There seems to be many variables influencing the plot, making it difficult to identify a general linear relationship

We only find linear behaviour when we check very specific cases. For example here, categorised by hours and only looking at the weekend we see a relationship which looks linear/transformable to linear.

Linear regression with covariate t1 and residuals

Hard to interpret Box-Cox transformation, $\hat{y} = [\hat{\lambda}(\hat{\beta}_0 + \hat{\beta}_1 x) + 1]^{1/\hat{\lambda}}$

